Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/13/2007
PI Name:	Moldover, Michael R. Ph.D.		
Project Title:	Acoustic Study of Critical Phenomena in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	michael.moldover@nist.gov	**Fax:**	FY 301 869-5924
PI Organization Type:	GOVERNMENT	**Phone:**	301 975-2459
Organization Name:	National Institute of Standards and Technology		
PI Address 1:	Code 8360		
PI Address 2:			
PI Web Page:			
City:	Gaithersburg	**State:**	MD
Zip Code:	20899-8360	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/21/2004	**End Date:**	04/20/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hall, Nancy Rabel	**Contact Phone:**	216-433-5643
Contact Email:	nancy.r.hall@nasa.gov		
Flight Program:			
Flight Assignment:	4/04 Note: grant renewed 4 yrs/new NAG#		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gillis, Keith (NIST) Shinder, Iosif (NIST)		
Grant/Contract No.:	NNC04IA20I		
Performance Goal No.:			
Performance Goal Text:			

	Using a double-Helmholtz acoustic resonator, also known as a Greenspan viscometer, we will measure the speed of sound and acoustic dissipation in a sample of xenon within 6 mK of the critical temperature. The speed of sound data will yield accurate measurement of the constant-volume heat capacity, and the quality factor of the resonator will be used to determine the bulk viscosity. These measurements will be extended to resonators made of different materials to test our understanding of thermoacoustic boundary layers. These measurements will be extended to off-critical densities to be certain that we obtained the correct critical density and to search for the predicted divergence of the second derivative of the chemical potential with respect to temperature. If the divergence is found, it will show that fluids do not have the symmetry of the Ising model in the usual thermodynamic variables.
Task Description:	The objective of this research is to measure the bulk viscosity of xenon near its liquid-vapor critical point using a novel acoustic resonator. The new measurements will be at frequencies that are a factor of 4000 lower than previous measurements, allowing us to extend the range of useful data a factor of 60 closer to the critical temperature. The bulk viscosity (or second viscosity) is the transport coefficient associated with dilational motion of a fluid. The bulk viscosity of a gas is often negligible in comparison to the shear viscosity. However, systems such as critical fluids or polyatomic gases may require many collisions before their internal degrees of freedom come into equilibrium with translational degrees of freedom, and the bulk viscosity can become much larger than the shear viscosity. Our measurements in microgravity will provide the most rigorous possible test of the theory for the bulk viscosity, i.e., the theory for the time-correlations of pairs of critical fluctuations. The quality-factor data will also test the theory of the thermal boundary layer in heretofore inaccessible ranges of frequency and fluid properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is fundamental; thus, its consequences are not immediate; however they are general. We now understand the bulk viscosity of xenon near its critical point. The bulk viscosity is the property that dominates the attenuation sound (and shock waves) in near critical fluids. We plan to extend this understanding to off-critical densities. The universality of critical phenomena quantitatively predicts the bulk viscosity of, for example, near-critical steam. Thus, we are contribution to the modeling and control of shock waves in turbines that use near-critical steam to generate electricity.
Task Progress:	The principle progress was in analyzing data, documenting the research and publishing it. The following papers were published in archival journals: Gillis, K. A., Shinder, I. I., and Moldover, M. R., “Bulk Viscosity of Stirred Xenon near the Critical Point”, Phys. Rev. E 72, 051201 (2005). Gillis, K. A., Shinder, I. I., and Moldover, M. R., “Bulk Viscosity, Thermoacoustic Boundary Layers, and Adsorption near the Critical Point”, Phys. Rev. Letters 97, 104502 (2006) The publication in Physical Review E filled twenty (20) journal pages and thoroughly documented the measurements and the acoustic model used to interpret the data very close to the critical point. The paper documented our technique for using forced convection to stir near critical xenon. With this novel technique, we were able to obtain high-quality speed-of-sound data and acoustic- attenuation data two decades closer to the critical point than previous investigations. Thus, we were able to simulate a microgravity experiment on earth. Despite the high-quality data and the rigorous analysis, a puzzle remained: Why did we fail to observe the predicted peak the acoustic losses in the thermoacoustic boundary layer. The prediction appears to be independent of the details of the acoustic model and details of the properties of xenon. All that is required is that both the thermal effusivity of xenon and the bulk viscosity of xenon increase strongly as the critical point is approached. The increasing effusivity increases the attenuation until the effusivity of xenon exceeds the effusivity of the metal walls of its container (the resonator). When this happens, the attenuation levels off. As the critical point is approached even more closely, the bulk viscosity continues to increase and effectively stiffens the thermo-acoustic boundary layer, thereby reducing the attenuation. In the Physical Review Letter, we explained the apparent inconsistency. We showed that critical adsorption of the xenon on the metal raised the density of the adsorbed layer above that of the bulk xenon and limited the approach of the adsorbed layer to the critical point. Thus, the acoustic resonator was filled with near-critical xenon; however, the boundary layer was further from the critical point because of its higher density. Contents of Publication in Preparation Prior to BVX, models for low-frequency acoustic resonance measurements in near-critical fluids were limited, in part, by an incomplete acoustic theory [1] that assumed low dissipation and neglected heat transport within the resonator wall. When a fluid is close to its liquid-vapor critical point, many thermophysical properties are extremely large, acoustic dissipation is large, and heat transport in solid boundaries is important. The theory described in Gillis, et al, [2 ,3] is based on the linearized Navier-Stokes equation, the continuity equation, the diffusion equation for heat transport (in the fluid and in the solid wall), and thermodynamic relationships among the fluctuating fields. The theory does not assume that the acoustic dissipation (near the boundaries and in the bulk fluid) is small. The solid is assumed to be rigid but able to support a thermal wave. The assumed boundary conditions are 1) the normal component of fluid velocity is zero at the wall (rigid walls), 2) the tangential component of the fluid velocity is zero at the wall (no slip), 3) the temperatures of the wall and the fluid are equal at the interface (local equilibrium), and 4) normal component of the heat flux across the interface is conserved (energy conservation). We used this extended theory, subject to the boundary conditions and the geometry, to develop a resonator model with which we analyzed our acoustic measurements in near-critical xenon. In a forthcoming publication, we derive expressions for the acoustic modes of a cylindrical chamber and of a double Helmholtz resonator that are valid for near-critical fluids and that include the thermophysical properties of the rigid resonator wall. [1] Theoretical Acoustics, P. M. Morse and K. U. Ingard, (McGraw-Hill, New York, 1968). [2] “Thermoacoustic boundary layers near the liquid-vapor critical point,” K. A. Gillis, I. I. Shinder, and M. R. Moldover, Phys. Rev. E 70, 021201 (2004). [3] “Bulk viscosity of stirred xenon near the critical point,” K. A. Gillis, I. I. Shinder, and M. R. Moldover, Phys. Rev. E

	72, 051201 (2005).
Bibliography Type:	Description: (Last Updated: 09/16/2010)
Articles in Peer-reviewed Journals	Gillis KA, Shinder II, Moldover MR. "Bulk viscosity, thermoacoustic boundary layers, and adsorption near the critical point of xenon." Phys Rev Lett. 2006 Sep 8;97(10):104502. PMID: 17025823 , Sep-2006
Articles in Peer-reviewed Journals	Gillis KA, Shinder II, Moldover MR. "Bulk viscosity of stirred xenon near the critical point." Phys Rev E Stat Nonlin Soft Matter Phys. 2005 Nov;72(5 Pt 1):051201. PMID: 16383593 , Nov-2005